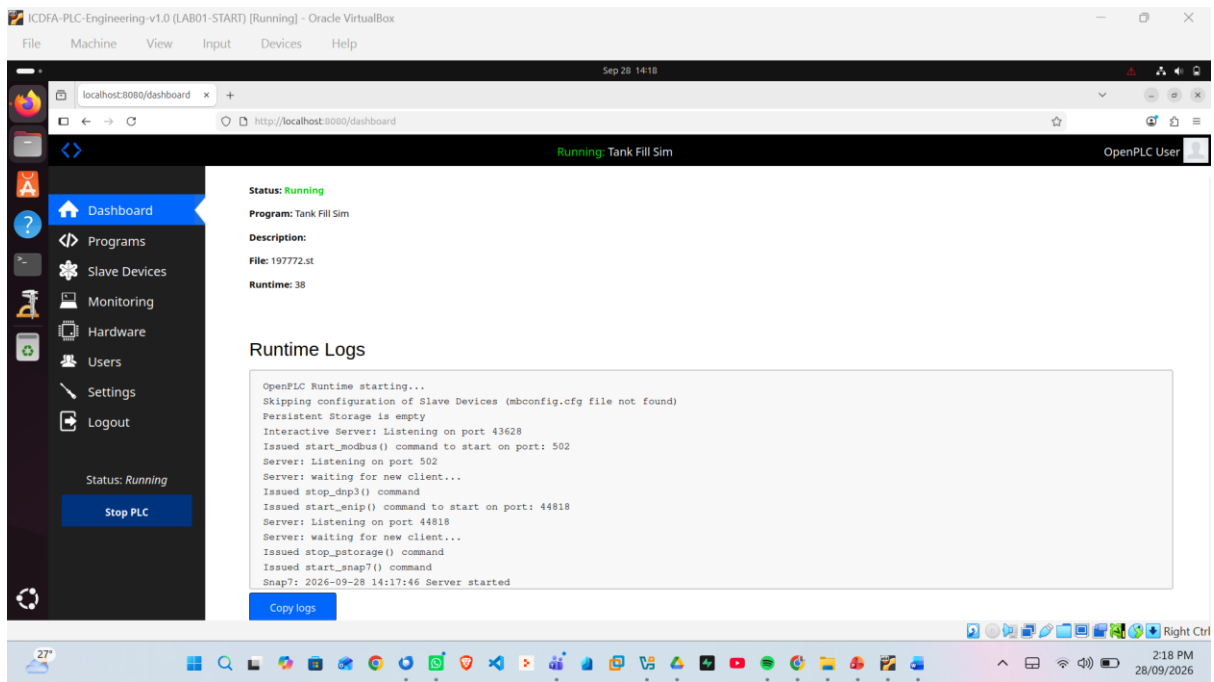
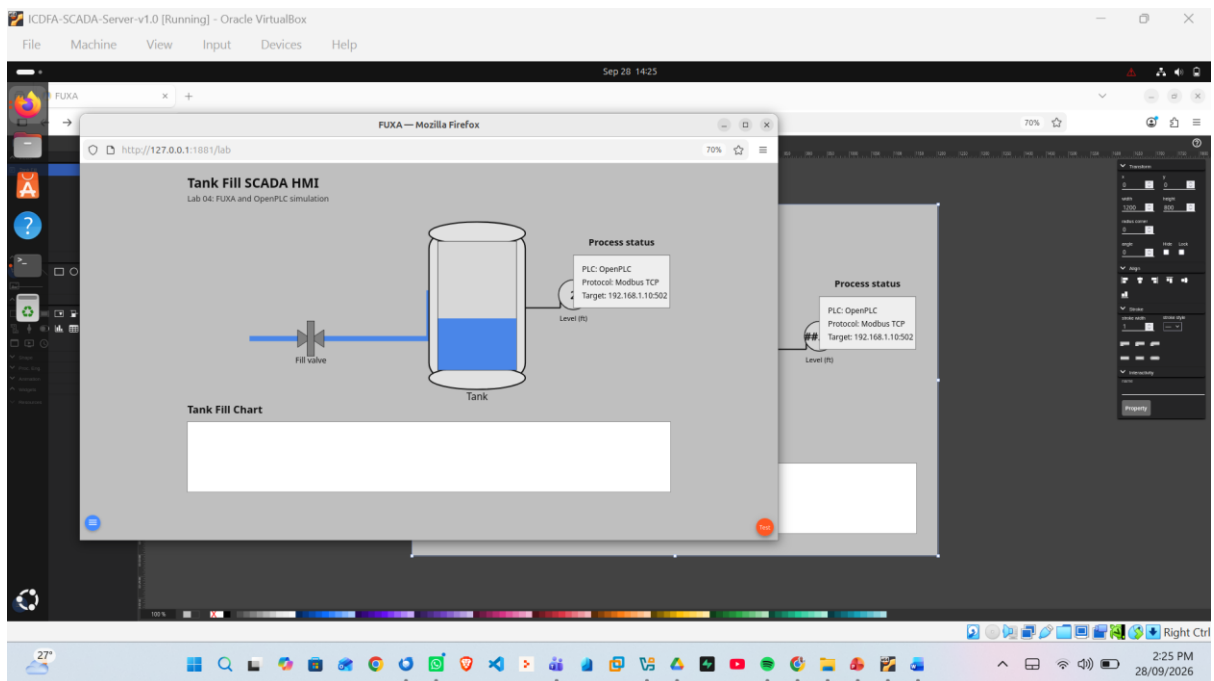
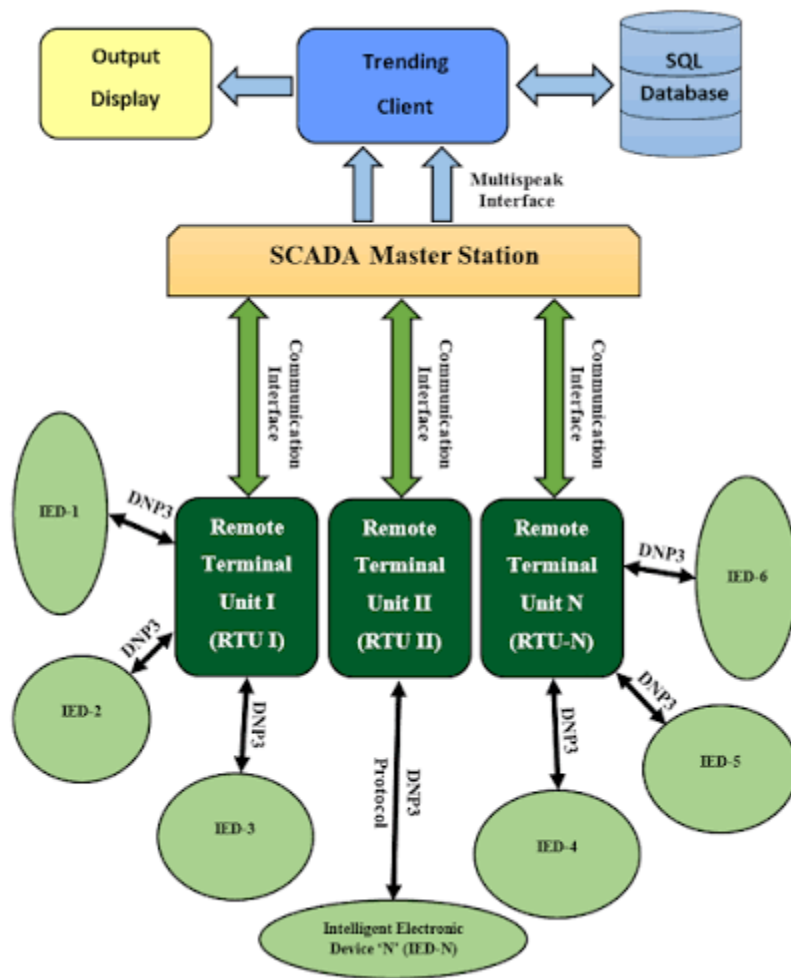




Screenshot showing OpenPLC Runtime in Running state.



Screenshot showing the FUXA tank-fill HMI in operation.



SCADA architecture diagram

Difference between PLC control logic and HMI/SCADA visualization

A PLC and an HMI/SCADA system perform different but connected roles in an industrial control system. The PLC is responsible for executing the control logic that operates the physical process. It receives signals from field inputs such as sensors, processes the programmed logic, and sends commands to outputs such as valves and pumps. In this lab, the OpenPLC Runtime executes the tank fill program and controls the simulated tank process.

The HMI/SCADA layer provides the operator with a graphical representation of the process. It displays process information, monitors inputs and outputs, allows operators to view system conditions, and provides supervisory control functions. The FUXA HMI used in this lab displays the tank, valve, level transmitter, and process trend information for operator awareness.

SCADA systems also provide additional functions such as alarming, event recording, and historical data storage. Alarms notify operators of abnormal conditions, event records capture operator or automation actions, and historized data allows process trends and analysis.

Therefore, the PLC performs the real-time automated control, while SCADA/HMI provides monitoring, visualization, data collection, and supervisory interaction with the process.

Knowledge Check

1. What is the primary purpose of an HMI in a SCADA system?

The primary purpose of an HMI is to provide a graphical interface that allows operators to view, monitor, and interact with industrial processes. It displays process information and the status of inputs and outputs through screens or dashboards.

2. What is the difference between an alarm, an event record and historized process data?

An alarm alerts operators about abnormal process conditions that require attention. An event record captures actions performed by operators or automation systems, such as starting a pump or turning on equipment. Historized process data stores process values over time and is used for trending, analysis, reporting, and reviewing past performance.

3. Why should a SCADA operator have context rather than only raw sensor values?

A SCADA operator needs context because raw sensor values alone do not provide enough understanding of the industrial process. Visualization through an HMI presents information in a meaningful way by showing process status, equipment relationships, trends, and conditions, allowing operators to understand and respond appropriately.

4. What role does the PLC play compared with the SCADA application?

The PLC performs the actual control of the industrial process by executing programmed logic and interacting with field inputs and outputs. The SCADA application provides supervisory monitoring, visualization, alarming, event recording, and historical data access. SCADA collects information from PLCs and displays it through an operator interface.